LAPORAN PRAKTIKUM

POSTTEST 1

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

Nama MEILONIE

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PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

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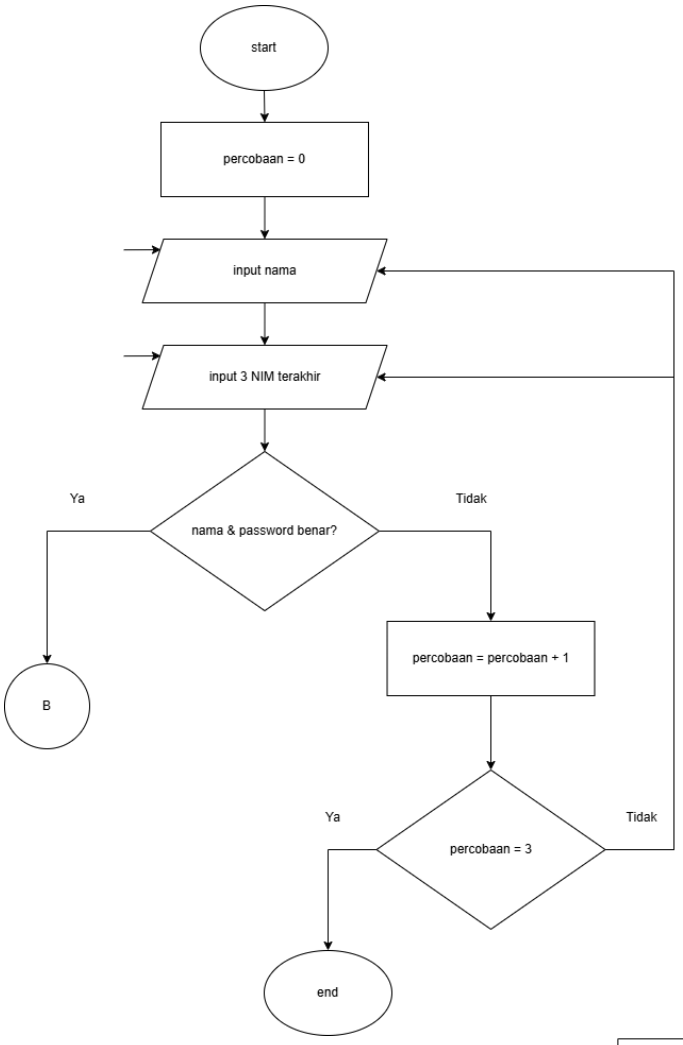
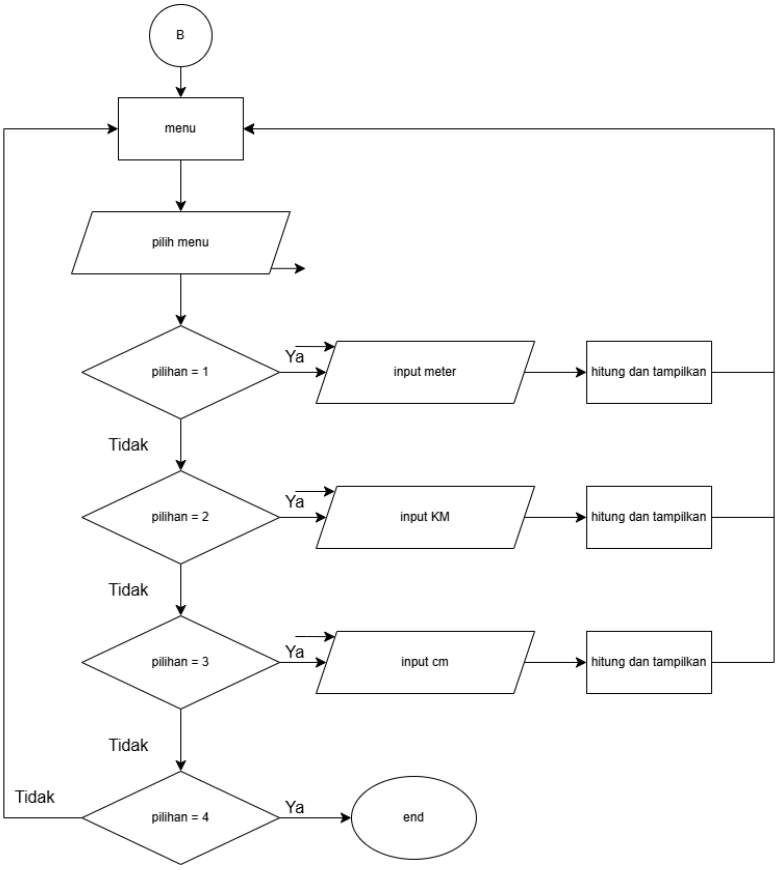
1. Flowchart


start
percobaan = 0
input nama
input 3 NIM terakhir
nama & password benar?
Ya
Tidak
B
percobaan = percobaan + 1
percobaan = 3
Ya
Tidak
end


B
menu
pilih menu
pilihan = 1
Ya
input meter
hitung dan tampilkan
Tidak
pilihan = 2
Ya
input KM
hitung dan tampilkan
Tidak
pilihan = 3
Ya
input cm
hitung dan tampilkan
Tidak
pilihan = 4
Ya
end
Tidak

2. Deskripsi Singkat Program

Untuk membuat sistem konversi satuan panjang (meter, kilometer, dan centimeter) yang hanya bisa diakses setelah user berhasil login, serta dapat digunakan berulang kali sampai user memilih keluar.

3. Source Code

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    string nama, password;
    int percobaan = 0;
    string nama_benar = "Meilonie";
    string password_benar = "128";

    while (percobaan < 3) {
        cout << "===================================\n";
        cout << "         LOGIN PROGRAM             \n";
        cout << "===================================\n";
        cout << "Masukkan Nama     : ";
        cin >> nama;
        cout << "Masukkan Password (3 digit NIM): ";
        cin >> password;

        if (nama == nama_benar && password == password_benar) {
            cout << "\nLogin Berhasil!\n";
            break;
        } else {
            percobaan++;
            cout << "\nLogin Gagal! Percobaan ke-" << percobaan << endl;
        }

        if (percobaan == 3) {
            cout << "\nAnda salah 3 kali. Program berhenti.\n";
            return 0;
        }
    }

    int pilihan;
    double nilai;

    do {
        cout << "\n===================================\n";
        cout << "        MENU KONVERSI PANJANG      \n";
        cout << "===================================\n";
        cout << "1. Meter ? Kilometer & Centimeter\n";
        cout << "2. Kilometer ? Meter & Centimeter\n";
        cout << "3. Centimeter ? Meter & Kilometer\n";
        cout << "4. Keluar\n";
        cout << "===================================\n";
        cout << "Pilih menu (1-4): ";
        cin >> pilihan;
```

```
48
49          switch (pilihan) {
50              case 1:
51                  cout << "\nMasukkan nilai dalam Meter: ";
52                  cin >> nilai;
53                  cout << fixed << setprecision(2);
54                  cout << "Hasil: \n";
55                  cout << nilai << " meter = "
56                       << nilai / 1000 << " kilometer\n";
57                  cout << nilai << " meter = "
58                       << nilai * 100 << " centimeter\n";
59                  break;
60
61              case 2:
62                  cout << "\nMasukkan nilai dalam Kilometer: ";
63                  cin >> nilai;
64                  cout << fixed << setprecision(2);
65                  cout << "Hasil: \n";
66                  cout << nilai << " kilometer = "
67                       << nilai * 1000 << " meter\n";
68                  cout << nilai << " kilometer = "
69                       << nilai * 100000 << " centimeter\n";
70                  break;
71
72              case 3:
73                  cout << "\nMasukkan nilai dalam Centimeter: ";
74                  cin >> nilai;
75                  cout << fixed << setprecision(2);
76                  cout << "Hasil: \n";
77                  cout << nilai << " centimeter = "
78                       << nilai / 100 << " meter\n";
79                  cout << nilai << " centimeter = "
80                       << nilai / 100000 << " kilometer\n";
81                  break;
82
83              case 4:
84                  cout << "\nTerima kasih telah menggunakan program ini\n";
85                  break;
86
87              default:
88                  cout << "\nPilihan tidak valid\n";
89          }
90
91      } while (pilihan != 4);
92
93      return 0;
94  }
```

4. Hasil Output

```
=====================================
          LOGIN PROGRAM
=====================================
Masukkan Nama     : Meilonie
Masukkan Password (3 digit NIM): 128

Login Berhasil!
```

1.1

```
=====================================
        MENU KONVERSI PANJANG
=====================================
1. Meter ? Kilometer & Centimeter
2. Kilometer ? Meter & Centimeter
3. Centimeter ? Meter & Kilometer
4. Keluar
=====================================
Pilih menu (1-4): 1

Masukkan nilai dalam Meter: 10
Hasil:
10.00 meter = 0.01 kilometer
10.00 meter = 1000.00 centimeter
```

1.2

```
=====================================
        MENU KONVERSI PANJANG
=====================================
1. Meter ? Kilometer & Centimeter
2. Kilometer ? Meter & Centimeter
3. Centimeter ? Meter & Kilometer
4. Keluar
=====================================
Pilih menu (1-4): 2

Masukkan nilai dalam Kilometer: 10
Hasil:
10.00 kilometer = 10000.00 meter
10.00 kilometer = 1000000.00 centimeter
```

1.3

```
=====================================
        MENU KONVERSI PANJANG
=====================================
1. Meter ? Kilometer & Centimeter
2. Kilometer ? Meter & Centimeter
3. Centimeter ? Meter & Kilometer
4. Keluar
=====================================
Pilih menu (1-4): 3

Masukkan nilai dalam Centimeter: 10
Hasil:
10.00 centimeter = 0.10 meter
10.00 centimeter = 0.00 kilometer
```

1.4

```
=====================================
        MENU KONVERSI PANJANG
=====================================
1. Meter ? Kilometer & Centimeter
2. Kilometer ? Meter & Centimeter
3. Centimeter ? Meter & Kilometer
4. Keluar
=====================================
Pilih menu (1-4): 4

Terima kasih telah menggunakan program ini!
```

1.5

5. Langkah-langkah GIT

5.1 GIT Add

```
PS D:\praktikum-apl> git init
Initialized empty Git repository in D:/praktikum-apl/.git/
```

Menambahkan perubahan file ke *staging area* (menyiapkan file untuk disimpan).

5.2 GIT Commit

```
PS D:\praktikum-apl> git commit -m "finish post test 1"
[master (root-commit) ab229fa] finish post test 1
 2 files changed, 94 insertions(+)
 create mode 100644 post-test/post-test-apl-1/postest-1-baru.cpp
 create mode 100644 post-test/post-test-apl-1/postest-1-baru.exe
```

git commit → Menyimpan perubahan dari *staging area* ke repository lokal dengan pesan commit.

5.3 GIT Push

```
PS D:\praktikum-apl> git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 447.91 KiB | 3.03 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/loniemei5-crypto/praktikum-apl.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS D:\praktikum-apl>
```

git push → Mengirim commit dari repository lokal ke repository remote